explore_entities_20261101

November 1, 2026

[1]:
```
import polars as pl
import seaborn_polars as snl
import re
```

[2]:
```
from tool_extraction import SEED, artifacts, artifact_siblings, repos,␣
↪mining_runs
```

[3]:
```
artifact_siblings.collect_schema()
```

[3]:
```
Schema([('repo_full_name', String),
        ('artifact_path', String),
        ('entry_name', String),
        ('entry_type', String),
        ('entry_size', Int64),
        ('entry_sha', String),
        ('content', String),
        ('content_fetched', Int64),
        ('skipped_reason', String)])
```

[4]:
```
(
    artifact_siblings
    .select("content_fetched", "content")
    .filter(pl.col("content_fetched") == 0)
    .select("content")
    .describe()
)
```

[4]:
```
shape: (9, 2)

 statistic    content
 ---          ---
 str          str

 count        0
 null_count   1036908
 mean         null
 std          null
 min          null
```

```
25%          null
50%          null
75%          null
max          null
```

[5]:
```
(
    artifact_siblings
    .select("content_fetched", "content")
    .filter(pl.col("content_fetched") == 1)
    .select("content")
    .describe()
)
```

[5]:
```
shape: (9, 2)

 statistic    content
 ---          ---
 str          str

 count        3497752
 null_count   10641
 mean         null
 std          null
 min
 25%          null
 50%          null
 75%          null
 max            New deep dive: The Future
              of…
```

[6]:
```
(
    artifact_siblings
    .select("content_fetched", "content")
    .filter(pl.col("content_fetched") == 2)
    .select("content")
    .describe()
)
```

[6]:
```
shape: (9, 2)

 statistic    content
 ---          ---
 str          str

 count        0
```

```
null_count   2719564
mean         null
std          null
min          null
25%          null
50%          null
75%          null
max          null
```

[7]:
```
labels = {0: "directory", 1: "text_stored", 2: "text_absent"}

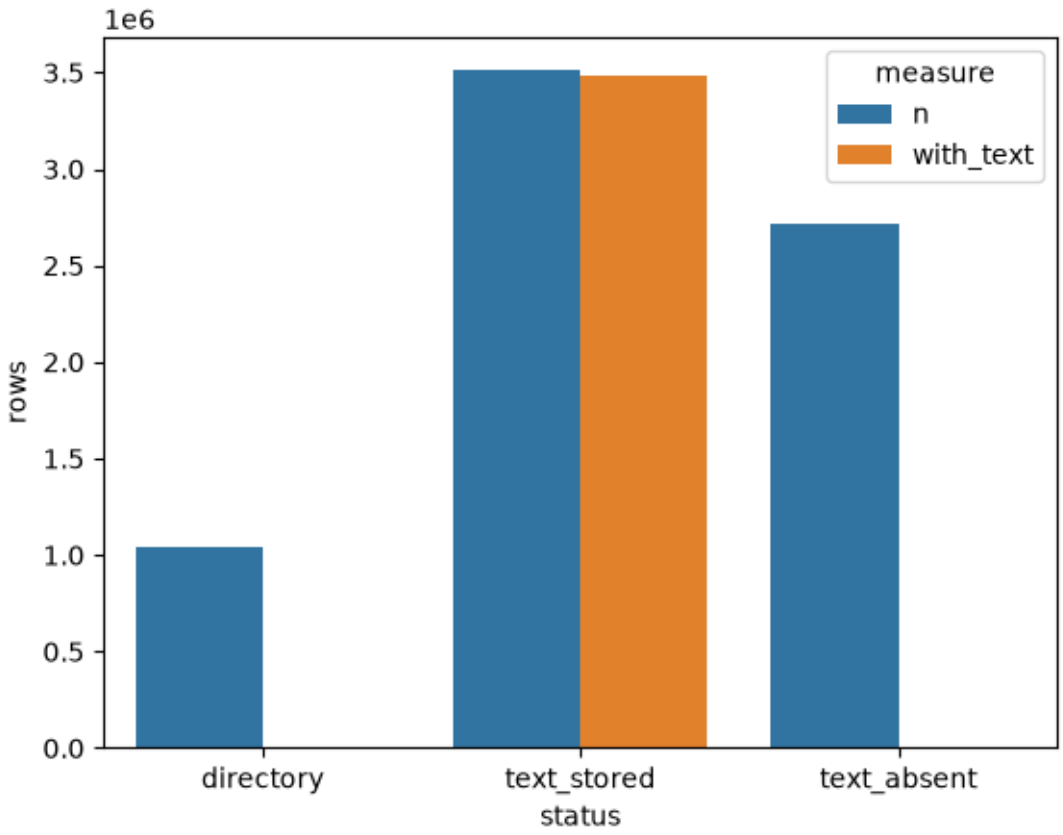
fetched = (
    artifact_siblings.group_by("content_fetched")
    .agg(
        pl.len().alias("n"),
        pl.col("content").str.strip_chars().replace("", None).is_not_null().
 ↪sum().alias("with_text"),
    )
    .with_columns(pl.col("content_fetched").replace_strict(labels).
 ↪alias("status"))
    .sort("content_fetched")
    .collect()
)
fetched.head()
```

[7]:
```
shape: (3, 4)

 content_fetched  n        with_text  status
 ---              ---      ---        ---
 i64              u32      u32        str

 0                1036908  0          directory
 1                3508393  3486179    text_stored
 2                2719564  0          text_absent
```

[8]:
```
plot_df = fetched.unpivot(
    on=["n", "with_text"],
    index="status",
    variable_name="measure",
    value_name="rows",
)
snl.barplot(plot_df, x="status", y="rows", hue="measure")
```



So only content_fetched == 1 has content, and even these has null values.

[9]:
```
artifact_siblings_with_content = (
    artifact_siblings
    .select("entry_sha", "content", "repo_full_name", "artifact_path",␣
 ↪"entry_name", "content_fetched")
    .filter(pl.col("content_fetched") == 1)
    .with_columns(pl.col("content").str.strip_chars().replace("", None))
    .filter(pl.col("content").is_not_null())
    .drop("content_fetched")
)
artifact_siblings_with_content.head().collect()
```

[9]:
```
shape: (5, 5)

 entry_sha           content             repo_full_name      artifact_path
 entry_name
 ---                 ---                 ---                 ---
 ---
```

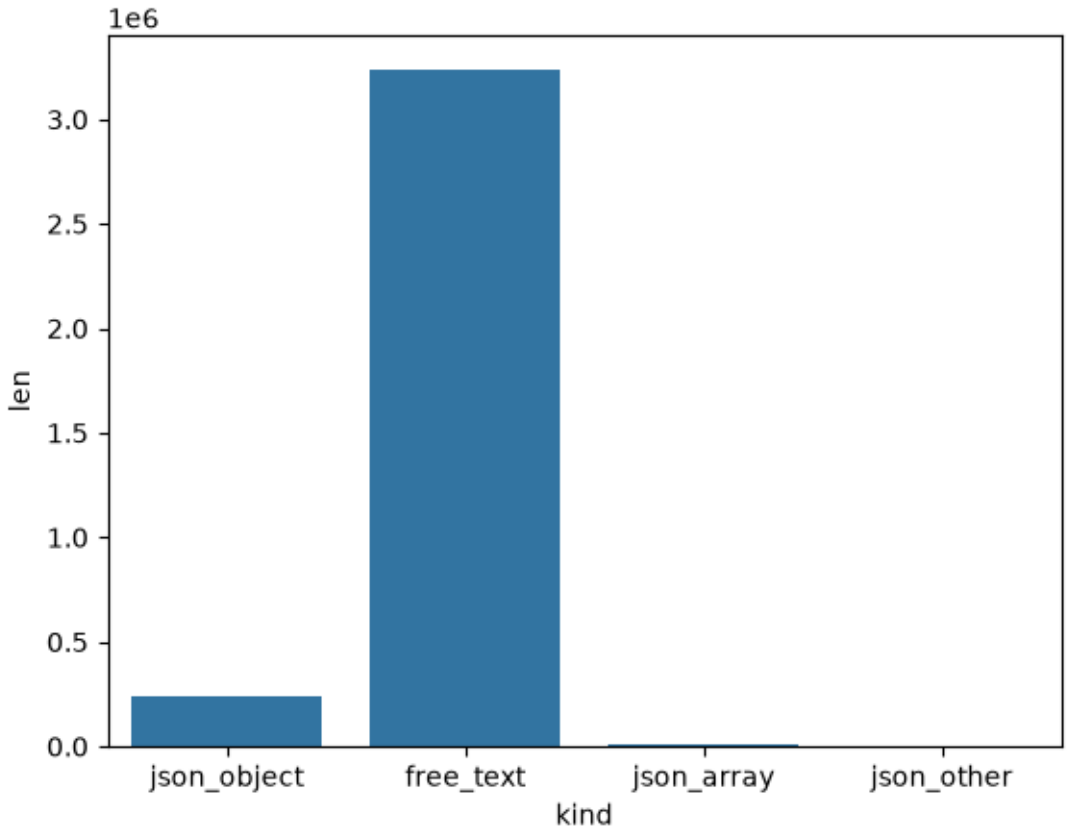
str	str	str	str
5043b7196d53053e4b1013403d3b64…	{ "version": 1, "id": "mid…	0-AI-UG/cate…	skills/cate-theme/SKILL.md
0cb2e2103def3abbf7a717668394c2…	{ "version": 1, "id": "pap…	0-AI-UG/cate…	skills/cate-theme/SKILL.md
f11253fd93ebee1109f0f1b9c7e44a…	{ "$schema": "http://json-sc…	0-AI-UG/cate…	skills/cate-theme/SKILL.md
ab25f27ebcecca37d6b26f26640da2…	# Coder Delegation Safety Har…	0-CYBERDYNE-SYSTE… MS-0/FFT_nano	skills/runtime/fft-coder-ops/S…
c92d9039aff7a8429…0926083326830…	#!/usr/bin/env bash set -euo p…	0-CYBERDYNE-SYSTE… MS-0/FFT_nano	skills/runtime/fft-debug/SKILL…

[10]:
```
content = pl.col("content")
valid = content.str.json_path_match("$").is_not_null()

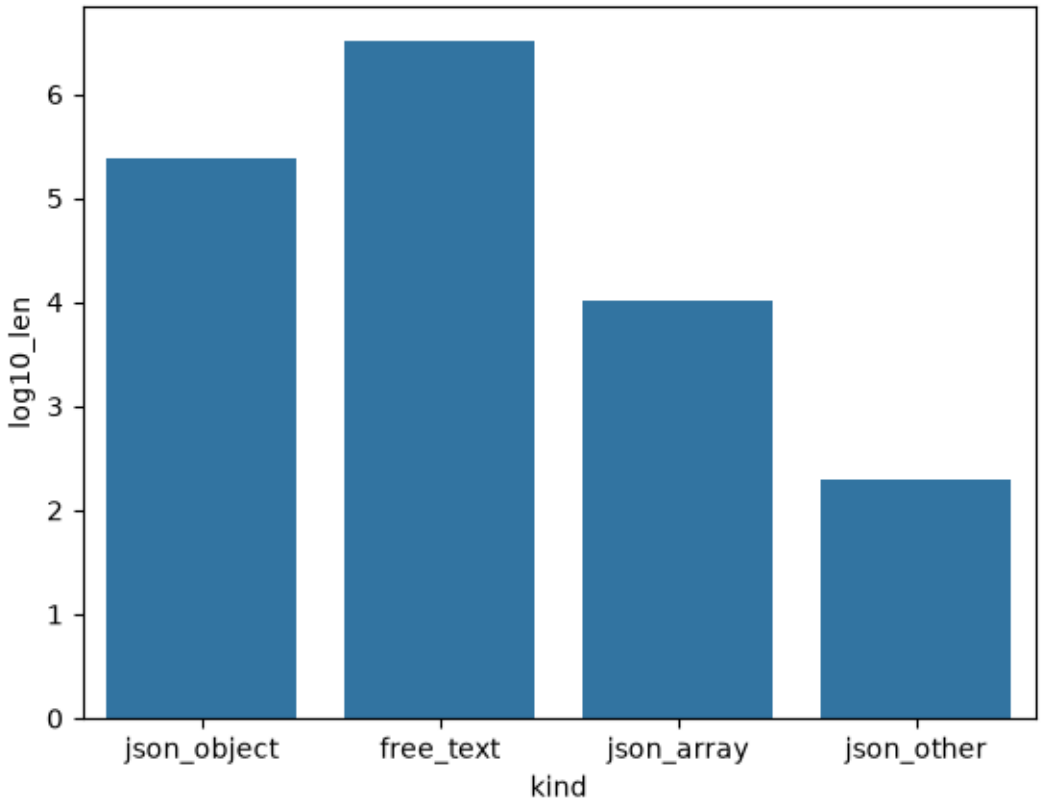
kinds = (
    artifact_siblings_with_content.select(
        pl.when(valid & content.str.starts_with("{")).then(pl.
 ↪lit("json_object"))
        .when(valid & content.str.starts_with("[")).then(pl.lit("json_array"))
```

```
        .when(valid).then(pl.lit("json_other"))
        .otherwise(pl.lit("free_text"))
        .alias("kind")
    )
    .group_by("kind")
    .len()
    .collect()
)
```

[11]:
```
snl.barplot(kinds, x="kind", y="len")
```



[12]:
```
plot = kinds.with_columns(pl.col("len").log10().alias("log10_len"))
snl.barplot(plot, x="kind", y="log10_len")
```



0.1 Gliner Entity

[13]:
```
from gliner import GLiNER

model = GLiNER.from_pretrained("urchade/gliner_small-v2.1")
```

```
/home/nichmorgan/Projects/nau/msr-data-
challenge-2027/scripts/.venv/lib/python3.13/site-
packages/torch/jit/_script.py:1491: FutureWarning: `torch.jit.script` is
deprecated. Please switch to `torch.compile` or `torch.export`.
  warnings.warn(
/home/nichmorgan/Projects/nau/msr-data-
challenge-2027/scripts/.venv/lib/python3.13/site-
packages/huggingface_hub/utils/_validators.py:190: UserWarning: The
`resume_download` argument is deprecated and ignored in `snapshot_download`.
Downloads always resume whenever possible.
  warnings.warn(
```

```
Downloading bytes:           |  0.00B
```

```
Reconstructing (incomplete total…): |          |  0.00B /  0.00B
```

```
Fetching 4 files:   0%|          | 0/4 [00:00<?, ?it/s]
```

[14]:
```
# Every column gets the same seed so the sampled rows stay aligned.
sample = artifact_siblings_with_content.select(pl.all().sample(n=500,␣
 ↪seed=SEED)).collect()
sample.head(3)
```

[14]:
```
shape: (3, 5)
```

entry_sha	content	repo_full_name	artifact_path	entry_name
str	str	str	str	str
69301b4ae010905ed…	#!/usr/bin/env python3 """Appl…	lightspeedwp/.git hub	agents/client-web site-discover…	scripts/comments _apply_patch.p…
0da363e9cb3014d24…	export function normalizeZulip…	niyazmft/openclaw -zulip-bridge	SKILL.md	src/normalize.ts
0d92ada056f8b2004…	"""Unit tests for linkedin.py …	yagyesh-bobde/xai -personalize-…	SKILL.md	tests/test_linke din.py

[15]:
```
MAX_LENGTH = 384
MAX_WORDS_CONTEXT = 5
HIT_THRESHOLD = 0.8
HIT_LABELS = ["tool", "organization"]
HIT_DTYPES = pl.Struct({
    "snippet": pl.List(pl.String),
    "entity": pl.String,
    "label": pl.String,
    "score": pl.Float64,
})

def get_snippet(text: str, entity: str) -> list[str]:
    pattern = re.compile(rf"(?<!\w){re.escape(entity)}(?!\w)")
    snippets = []
```

```
    for match in pattern.finditer(text):
        before = text[: match.start()].split()
        after = text[match.end() :].split()
        highlighted = f"[{match.group()}]"
        snippet = " ".join(before[-MAX_WORDS_CONTEXT:] + [highlighted] + after[:
 ↪MAX_WORDS_CONTEXT])
        if len(before) > MAX_WORDS_CONTEXT:
            snippet = "..." + snippet
        if len(after) > MAX_WORDS_CONTEXT:
            snippet = snippet + "..."
        snippets.append(snippet)
    return snippets

def extract(text: str) -> dict:
    clipped = (text or "")[:MAX_LENGTH]
    return [
        {
            "entity": ent["text"],
            "label": ent["label"],
            "score": ent["score"],
            "snippet": get_snippet(clipped, ent["text"]),
        }
        for ent in model.predict_entities(clipped, HIT_LABELS,␣
 ↪threshold=HIT_THRESHOLD)
    ]

hits = (
    sample.lazy()
    .select("entry_sha", "content")
    .with_columns(
        pl.col("content")
            .map_elements(extract, return_dtype=pl.List(HIT_DTYPES))
            .alias("hits")
    )
    .explode("hits")
    .unnest("hits")
    .select("entry_sha", "snippet", "entity", "label", "score")
)

with pl.Config(fmt_str_lengths=20_000):
    display(hits.head(20).collect())
```

```
shape: (20, 5)
```

entry_sha	snippet	entity	label	score
---	---	---	---	---

```
str                              list[str]                    str            str
↪      f64

0d92ada056f8b2004a03ec48e5   ["""Unit tests for          linkedin
↪organization   0.98536
3cbd1f068a4dae               [linkedin] .py pure
↪
                             helpers. Run with…",
↪
                             "…import sys
↪
                             sys.path.insert(0, os.path
↪
                             .dirname(os.path.dirname(_
↪
                             _file__))) import
↪
                             [linkedin] as L # noqa:
↪
                             E402…"]
↪
0d92ada056f8b2004a03ec48e5   ["…linkedin.py pure       pytest         tool
↪     0.9485
3cbd1f068a4dae               helpers. Run with [pytest]
↪
                             (`python3 -m pytest
↪
                             tests/test_linkedin.py
↪
                             -v`)…", "…Run with
↪
                             pytest (`python3 -m
↪
                             [pytest]
↪
                             tests/test_linkedin.py
↪
                             -v`) or directly
↪
                             (`python3…", "…or
↪
                             directly (`python3
↪
                             tests/test_linkedin.py`)
↪
```

```
                            when [pytest] is                                  ␣
↪
                            unavailable. """ import                           ␣
↪
                            os…"]                                             ␣
↪
0d92ada056f8b2004a03ec48e5  ["…linkedin.py pure       pytest        tool      ␣
↪    0.946572
3cbd1f068a4dae              helpers. Run with [pytest]                        ␣
↪
                            (`python3 -m pytest                               ␣
↪
                            tests/test_linkedin.py                            ␣
↪
                            -v`)…", "…Run with                                ␣
↪
                            pytest (`python3 -m                               ␣
↪
                            [pytest]                                          ␣
↪
                            tests/test_linkedin.py                            ␣
↪
                            -v`) or directly                                  ␣
↪
                            (`python3…", "…or                                 ␣
↪
                            directly (`python3                                ␣
↪
                            tests/test_linkedin.py`)                          ␣
↪
                            when [pytest] is                                  ␣
↪
                            unavailable. """ import                           ␣
↪
                            os…"]                                             ␣
↪
0d92ada056f8b2004a03ec48e5  ["…linkedin.py pure       pytest        tool      ␣
↪    0.936272
3cbd1f068a4dae              helpers. Run with [pytest]                        ␣
↪
                            (`python3 -m pytest                               ␣
↪
                            tests/test_linkedin.py                            ␣
↪
                            -v`)…", "…Run with                                ␣
↪
```

```
                          pytest (`python3 -m                          ␣
↪
                          [pytest]                                     ␣
↪
                          tests/test_linkedin.py                       ␣
↪
                          -v`) or directly                             ␣
↪
                          (`python3…", "…or                            ␣
↪
                          directly (`python3                           ␣
↪
                          tests/test_linkedin.py`)                     ␣
↪
                          when [pytest] is                             ␣
↪
                          unavailable. """ import                      ␣
↪
                          os…"]                                        ␣
↪
0d92ada056f8b2004a03ec48e5  ["""Unit tests for         linkedin      ␣
↪organization   0.983199
3cbd1f068a4dae              [linkedin] .py pure                        ␣
↪
                          helpers. Run with…",                         ␣
↪
                          "…import sys                                 ␣
↪
                          sys.path.insert(0, os.path                   ␣
↪
                          .dirname(os.path.dirname(_                   ␣
↪
                          _file__))) import                            ␣
↪
                          [linkedin] as L # noqa:                      ␣
↪
                          E402…"]                                      ␣
↪
…                         …                          …          …      ␣
↪       …
e674f7086fa7426e60d8d3823c  ["#!/bin/bash #            meeting-prep   tool  ␣
↪       0.858133
74ce6cfa83ccde              [meeting-prep]                             ␣
↪
                          /scripts/prep.sh -                           ␣
↪
```

```
                                Research attendees                                ␣
↪
                                for…", "…CONFIG & ARGS                            ␣
↪
                                --- PREP_DIR="${PREP_DIR:-                        ␣
↪
                                $HOME/.config/                                    ␣
↪
                                [meeting-prep] }" CONFIG_F                        ␣
↪
                                ILE="$PREP_DIR/config.json                        ␣
↪
                                " HISTORY_FILE="$PREP_DIR/                        ␣
↪
                                brief-history.json" LOG_FI                        ␣
↪
                                LE="$PREP_DIR/prep-log.jso                        ␣
↪
                                n" BRIEFS_DIR="$PREP_DIR/b                        ␣
↪
                                riefs"…"]                                         ␣
↪
e674f7086fa7426e60d8d3823c      ["#!/bin/bash #             meeting-prep   tool   ␣
↪       0.809411
74ce6cfa83ccde                  [meeting-prep]                                    ␣
↪
                                /scripts/prep.sh -                                ␣
↪
                                Research attendees                                ␣
↪
                                for…", "…CONFIG & ARGS                            ␣
↪
                                --- PREP_DIR="${PREP_DIR:-                        ␣
↪
                                $HOME/.config/                                    ␣
↪
                                [meeting-prep] }" CONFIG_F                        ␣
↪
                                ILE="$PREP_DIR/config.json                        ␣
↪
                                " HISTORY_FILE="$PREP_DIR/                        ␣
↪
                                brief-history.json" LOG_FI                        ␣
↪
                                LE="$PREP_DIR/prep-log.jso                        ␣
↪
```

```
                          n" BRIEFS_DIR="$PREP_DIR/b                                  ␣
↪
                          riefs"…"]                                                   ␣
↪
bf48e8b8fec43bc4b6fc6ee326 ["…""" Subdomain          Subfinder     tool              ␣
↪    0.917386
dcf4fffe1eaea8             Enumeration Pipeline with                                ␣
↪
                          [Subfinder] Automates                                       ␣
↪
                          subdomain discovery,                                        ␣
↪
                          validation, and…"]                                          ␣
↪
bf48e8b8fec43bc4b6fc6ee326 ["…= False) -> list:      subfinder     tool              ␣
↪    0.919183
dcf4fffe1eaea8             """Run [subfinder] against                               ␣
↪
                          a target dom"]                                              ␣
↪
17cb6801c9fd232fedd4e553b5 ["{ "source": "https://     github.com    ␣
↪organization   0.803482
d5937ff39f25b5             [github.com]                                             ␣
↪
                          /mattpocock/skills.git",                                    ␣
↪
                          "sourceType": "git",                                        ␣
↪
                          "repoUrl": "https://github                                  ␣
↪
                          .com/mattpocock/skills.git                                  ␣
↪
                          ",…", "…"https://githu                                      ␣
↪
                          b.com/mattpocock/skills.gi                                  ␣
↪
                          t", "sourceType": "git",                                    ␣
↪
                          "repoUrl": "https://                                        ␣
↪
                          [github.com]                                                ␣
↪
                          /mattpocock/skills.git",                                    ␣
↪
                          "subpath":                                                  ␣
↪
```

```
                                "migrate-to-shoehorn",                          ␣
↪
                                "installedAt": "2026-04-14                      ␣
↪
                                T10:06:51.370Z"…"]                              ␣
↪
```

[16]:
```
top = (
    hits.with_columns(pl.col("entity").str.to_lowercase().alias("key"))
    .group_by("key")
    .agg(
        pl.col("entry_sha").n_unique().alias("skills"),
        pl.col("label").mode().sort().first().alias("label"),
        pl.col("entity").mode().sort().first().alias("entity"),
    )
    .filter(pl.col("skills") >= 2)
    .sort(["skills", "key"], descending=[True, False])
    .collect()
)
snl.barplot(top, y="entity", x="skills", hue="label", order=top["entity"].
 ↪to_list())
```

